

DAY 02 – NETWORKING REPORT

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THEORY

1. What is a computer network?

A computer network is a group of connected computers and devices that communicate and share data, resources, and services.

2. What is the difference between a client and a server?

A client requests a service or resource, while a server provides the requested service or resource to clients.

3. What is a LAN?

LAN (Local Area Network) connects devices within a small geographical area such as a home, office, laboratory, or school.

4. What is a WAN?

WAN (Wide Area Network) connects networks across large geographical areas, such as cities, countries, or continents. The Internet is the largest example.

5. What is the purpose of a switch?

A switch connects devices within a LAN and forwards data frames to the appropriate device using MAC addresses.

6. What is the purpose of a router?

A router connects different networks and forwards IP packets between them. It commonly provides the path from a local network to the Internet.

7. What is a firewall?

A firewall is a security system that monitors and controls network traffic according to predefined security rules.

8. What is an IP address?

An IP address is a logical address assigned to a device on a network so that it can be identified and communicate using the Internet Protocol.

9. What is a MAC address?

A MAC address is a hardware-level address associated with a network interface. It is used for communication within a local network.

10. What is a default gateway?

A default gateway is usually the router address that a device uses to send traffic to destinations outside its local network.

11. What is DNS?

DNS (Domain Name System) translates human-readable domain names, such as example.com, into IP addresses that computers can use.

12. Why is networking knowledge important for cybersecurity?

Networking knowledge is important for cybersecurity because security professionals need to understand IP addresses, ports, protocols, traffic, and communication between devices. It helps in detecting attacks, analyzing network traffic, securing systems, and troubleshooting security incidents.

YOUR NETWORK INFORMATION

Question	Answer
13. Private IPv4 Address	10.84.163.114
14. Default Gateway	10.84.163.135

15. DNS Server	10.84.163.135
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Linux Command Note: My computer runs **Ubuntu Linux**, not Windows. Therefore, Linux equivalents were used for the requested Windows commands. **ip addr** and **ip route** were used for network information, **resolvectl status** for DNS, **ping** for connectivity, and **traceroute** as the Linux equivalent of Windows **tracert**.

NETWORK COMMAND SCREENSHOTS

1. IP configuration — ip addr

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether e8:d8:d1:ec:91:9e brd ff:ff:ff:ff:ff:ff
3: wlp0s20f3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether c8:09:a8:e6:1e:f5 brd ff:ff:ff:ff:ff:ff
    inet 10.84.163.114/24 brd 10.84.163.255 scope global dynamic noprefixroute wlp0s20f3
        valid_lft 2491sec preferred_lft 2491sec
    inet6 2409:40e3:5056:7da1:eea1:1382:fef1:396d/64 scope global temporary dynamic
        valid_lft 7173sec preferred_lft 7173sec
    inet6 2409:40e3:5056:7da1:1e42:32e7:aa2d:2e12/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 7173sec preferred_lft 7173sec
    inet6 fe80::e142:eb01:8f64:4d7d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: wlx00e02691d31b: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:e0:26:91:d3:1b brd ff:ff:ff:ff:ff:ff
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ hostname -I
```

2. Routing — hostname -I / ip route

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ hostname -I
10.84.163.114 2409:40e3:5056:7da1:eea1:1382:fef1:396d 2409:40e3:5056:7da1:1e42:32e7:aa2d:2e12
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip route
default via 10.84.163.135 dev wlp0s20f3 proto dhcp src 10.84.163.114 metric 600
10.84.163.0/24 dev wlp0s20f3 proto kernel scope link src 10.84.163.114 metric 600
```

3. DNS — resolvectl status

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ resolvectl status
Global
  Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (enp0s31f6)
  Current Scopes: none
  Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported

Link 3 (wlp0s20f3)
  Current Scopes: DNS
  Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Current DNS Server: 2409:40e3:5056:7da1::f7
  DNS Servers: 10.84.163.135 2409:40e3:5056:7da1::f7

Link 4 (wlx00e02691d31b)
  Current Scopes: none
  Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
```

SCREENSHOTS (CONTINUED)

4. Ping — ping google.com

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping google.com
PING google.com (2404:6800:4002:817::200e) 56 data bytes
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=1 ttl=115 time=36.0 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=2 ttl=115 time=55.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=3 ttl=115 time=57.5 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=4 ttl=115 time=51.5 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=5 ttl=115 time=38.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=6 ttl=115 time=70.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=7 ttl=115 time=65.8 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=8 ttl=115 time=78.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=9 ttl=115 time=62.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=10 ttl=115 time=52.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=11 ttl=115 time=57.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=12 ttl=115 time=48.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=13 ttl=115 time=53.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=14 ttl=115 time=52.1 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=15 ttl=115 time=55.8 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=16 ttl=115 time=51.3 ms
^C
--- google.com ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15005ms
rtt min/avg/max/mdev = 35.955/55.365/78.217/10.264 ms
```

5. Traceroute — traceroute google.com (Linux equivalent of tracert)

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ traceroute google.com
traceroute to google.com (142.250.182.14), 30 hops max, 60 byte packets
 1 _gateway (10.84.163.135) 2.302 ms 3.387 ms 4.262 ms
 2 * * *
 3 255.0.0.2 (255.0.0.2) 51.457 ms 54.517 ms 53.131 ms
 4 255.0.0.3 (255.0.0.3) 53.089 ms 50.498 ms 52.928 ms
 5 192.168.227.98 (192.168.227.98) 48.761 ms 192.168.227.99 (192.168.227.99) 49.272 ms 192.168.227.98 (192.168.227.98) 48.761 ms
 6 172.25.86.124 (172.25.86.124) 52.768 ms 172.25.119.228 (172.25.119.228) 31.816 ms 172.25.86.126 (172.25.86.126) 31.816 ms
 7 * * *
 8 * * *
 9 72.14.195.34 (72.14.195.34) 49.340 ms * *
10 192.178.83.245 (192.178.83.245) 48.374 ms 48.813 ms 31.347 ms
11 209.85.252.65 (209.85.252.65) 42.114 ms 41.808 ms 209.85.252.71 (209.85.252.71) 41.326 ms
12 t2delb-bf-in-f14.1e100.net (142.250.182.14) 40.113 ms 209.85.252.71 (209.85.252.71) 39.360 ms 209.85.252.65 (209.85.252.65) 41.326 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$
```